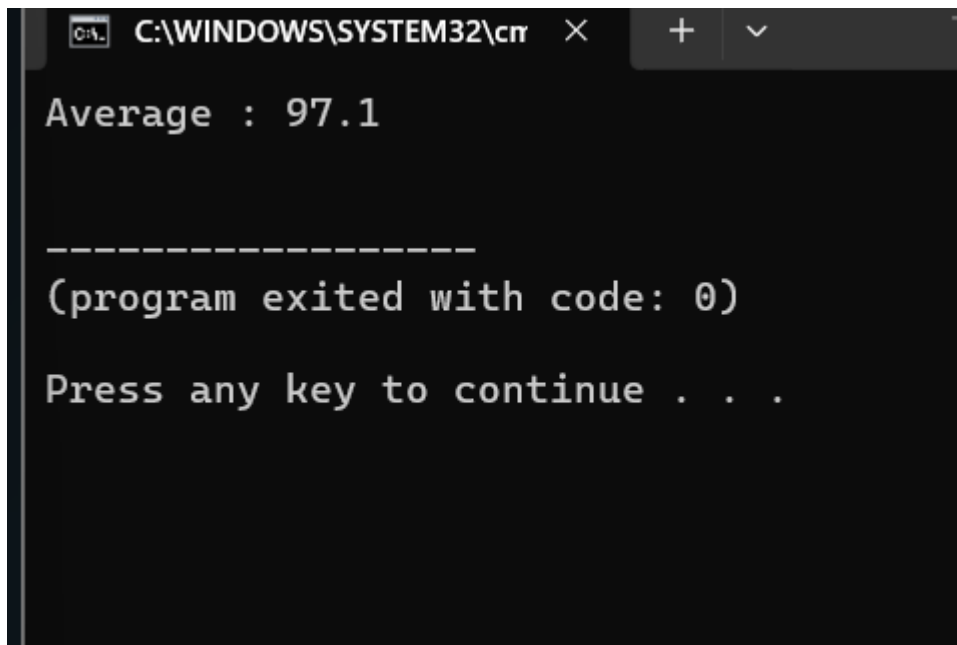


01.

```
class Demo {  
    public static void main(String args[]){  
        int tot = 971;  
        double avg;  
        avg = (double) tot/10; A. avg = (double) tot/10;  
        System.out.println("Average : " + avg);  
    }  
}
```

Output



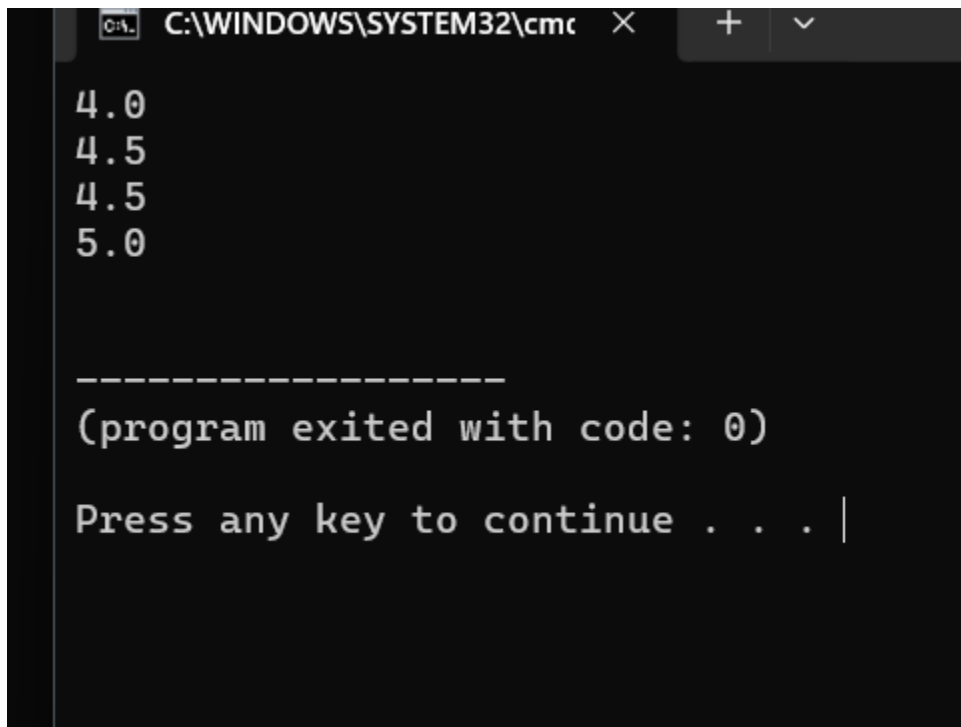
A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\WINDOWS\SYSTEM32\cmd' and standard window controls. The command prompt displays the output of a Java program: 'Average : 97.1'. Below this, there is a line of dashes '-----' followed by the message '(program exited with code: 0)'. At the bottom, it says 'Press any key to continue . . .'. The text is displayed in a white monospaced font on a black background.

02.

Code.

```
class Example {  
    public static void main (String args[]){  
        double d;  
        d=5/2+5/2;  
        System.out.println(d);  
        d=5/2.0+5/2;  
        System.out.println(d);  
        d=5/2+5.0/2;  
        System.out.println(d);  
        d=5/2.0+5/2.0;  
        System.out.println(d);  
    }  
}
```

output



```
C:\WINDOWS\SYSTEM32\cmd.exe X + v  
4.0  
4.5  
4.5  
5.0  
  
-----  
(program exited with code: 0)  
Press any key to continue . . . |
```

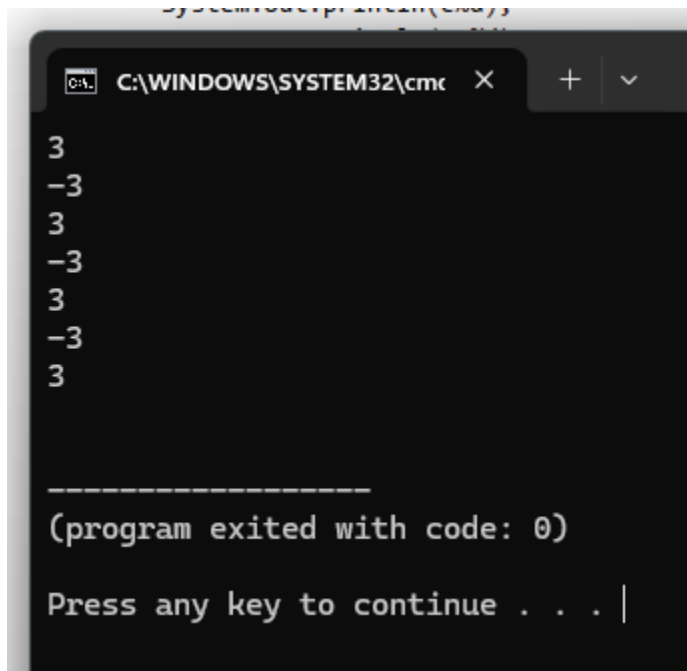
Answer - E

03.

Code.

```
class Example{  
    public static void main(String args[]){  
        int a=10, b=7,c=-10,d=-7;  
        System.out.println(a%b);  
        System.out.println(-a%b);  
        System.out.println(a%-b);  
        System.out.println(-a%-b);  
        System.out.println(+a%+b);  
        System.out.println(c%d);  
        System.out.println(-c%d);  
    }  
}
```

Output.



A screenshot of a Windows command prompt window. The title bar shows the path 'C:\WINDOWS\SYSTEM32\cmd.exe'. The window has a dark background with white text. The output of the Java program is displayed as follows:

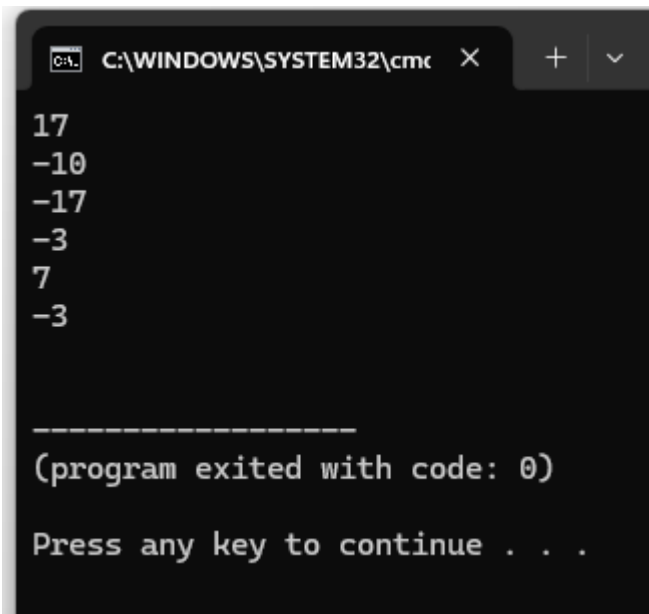
```
3  
-3  
3  
-3  
3  
-3  
3  
  
-----  
(program exited with code: 0)  
Press any key to continue . . . |
```

04.

Code.

```
class Example{  
    public static void main(String args[]){  
        int x=10,y=7;  
        System.out.println(x+y);  
        System.out.println(-x);  
        System.out.println(-x-y);  
        System.out.println(-(x-y));  
        System.out.println(+y);  
        System.out.println(+y-x);  
    }  
}
```

Output



A screenshot of a Windows command prompt window. The title bar shows the file path "C:\WINDOWS\SYSTEM32\cmd.exe" and standard window controls. The command prompt displays the output of a Java program: the numbers 17, -10, -17, -3, 7, and -3 are printed on separate lines. Below these numbers is a dashed line, followed by the text "(program exited with code: 0)". At the bottom, it says "Press any key to continue . . .".

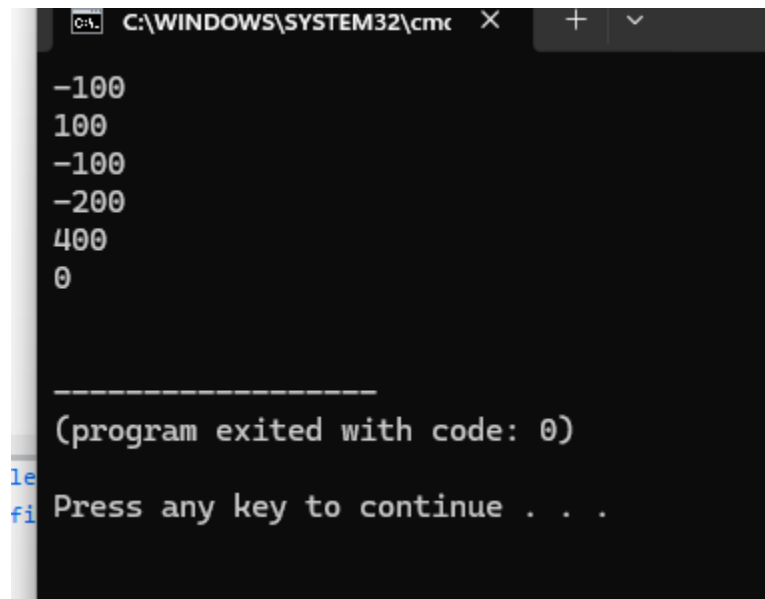
```
C:\WINDOWS\SYSTEM32\cmd.exe X + v  
17  
-10  
-17  
-3  
7  
-3  
  
-----  
(program exited with code: 0)  
Press any key to continue . . .
```

05.

Code.

```
class Example{
    public static void main(String args[]){
        int x = -100;
        x = +x;
        System.out.println(x);
        x = -x;
        System.out.println(x);
        x = -x;
        System.out.println(x);
        x = x + x;
        System.out.println(x);
        x = -x - x;
        System.out.println(x);
        x = x - x;
        System.out.println(x);
    }
}
```

Output.



A screenshot of a Windows command prompt window. The title bar shows the path 'C:\WINDOWS\SYSTEM32\cmd' and standard window controls. The command prompt displays the output of the Java program: -100, 100, -100, -200, 400, 0. Below the output, a separator line is shown, followed by the text '(program exited with code: 0)'. At the bottom, it says 'Press any key to continue . . .'. On the left side of the image, there is a vertical text label 'le' above 'fi'.

```
C:\WINDOWS\SYSTEM32\cmd X + v
-100
100
-100
-200
400
0

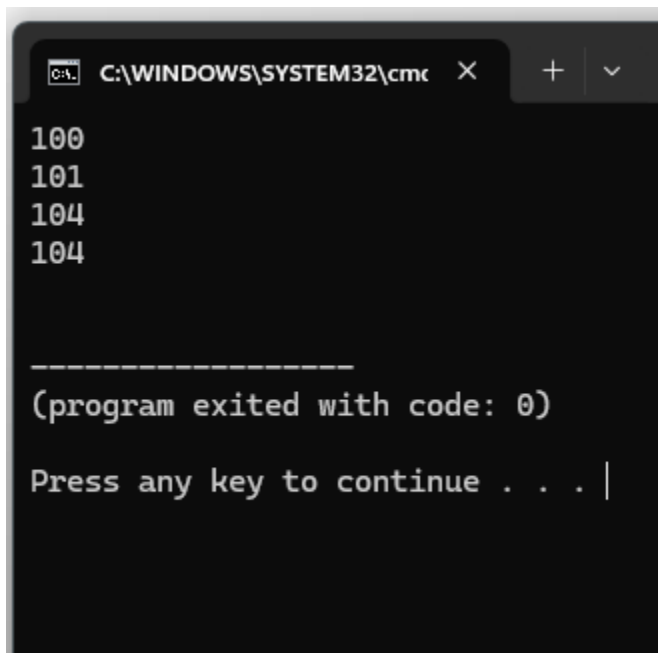
-----
(program exited with code: 0)
Press any key to continue . . .
```

06.

Code.

```
class Example{  
    public static void main(String args[]){  
        int x=100;  
        System.out.println(x++);  
        System.out.println(x++);  
        x++;  
        System.out.println(++x);  
        System.out.println(x++);  
    }  
}
```

Output.



The screenshot shows a Windows command prompt window with the title bar "C:\WINDOWS\SYSTEM32\cmd.exe". The output of the Java program is displayed as follows:

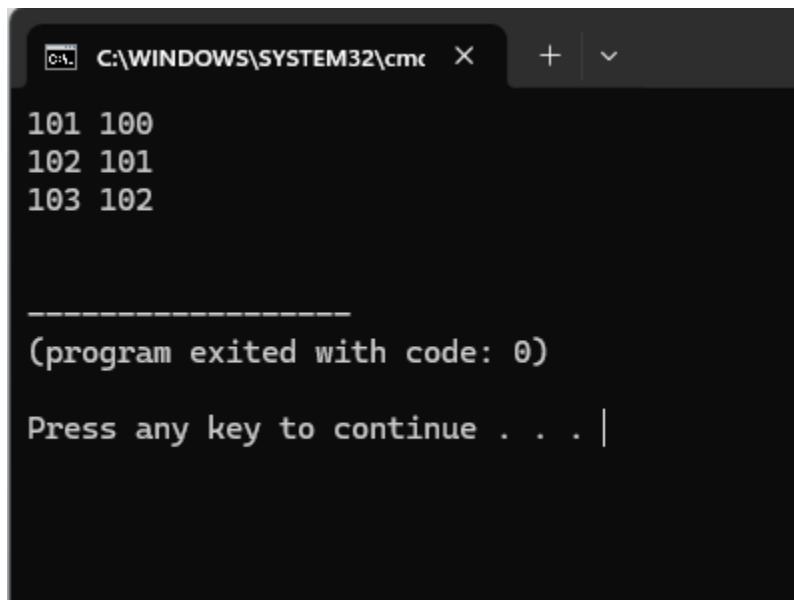
```
100  
101  
104  
104  
  
-----  
(program exited with code: 0)  
Press any key to continue . . . |
```

07.

Code.

```
class Example{  
    public static void main(String args[]){  
        int x=100,y;  
        y=x++;  
        System.out.println(x+" "+y);  
        y=x++;  
        System.out.println(x+" "+y);  
        y=x++;  
        System.out.println(x+" "+y);  
    }  
}
```

Output.



A screenshot of a Windows command prompt window. The title bar shows the path 'C:\WINDOWS\SYSTEM32\cmd' and standard window controls. The command prompt displays the output of the Java program: three lines of space-separated integers '101 100', '102 101', and '103 102'. Below these, a separator line of dashes is shown, followed by the message '(program exited with code: 0)'. At the bottom, the prompt 'Press any key to continue . . . |' is visible with a cursor.

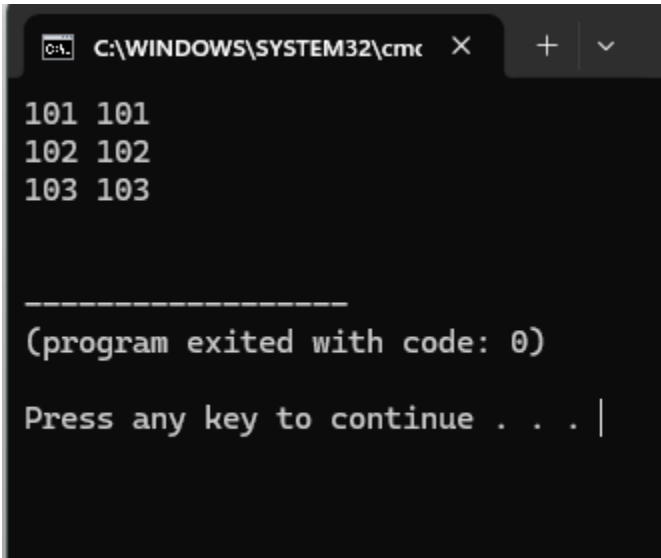
```
C:\WINDOWS\SYSTEM32\cmd X + v  
101 100  
102 101  
103 102  
  
-----  
(program exited with code: 0)  
Press any key to continue . . . |
```

08.

Code.

```
class Example{
    public static void main(String args[]){
        int x=100,y;
        y=++x;
        System.out.println(x+" "+y);
        y=++x;
        System.out.println(x+" "+y);
        y=++x;
        System.out.println(x+" "+y);
    }
}
```

Output.

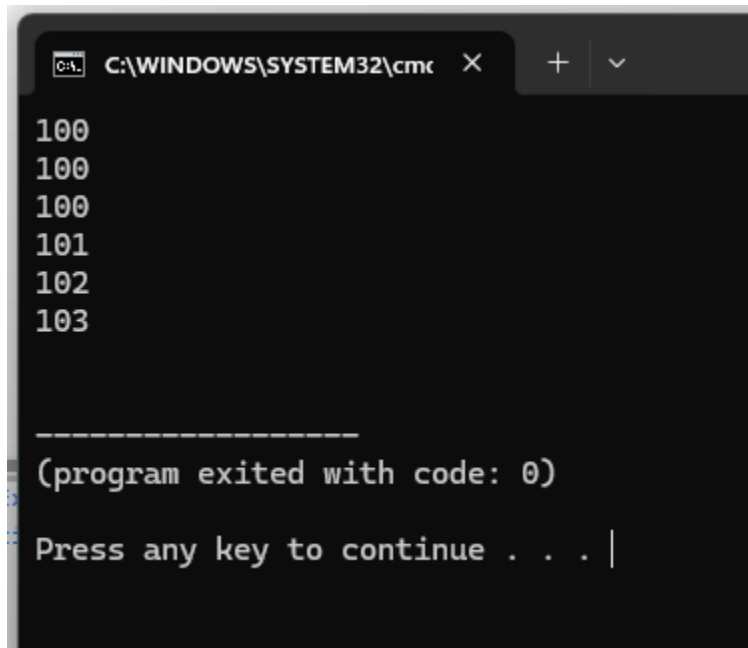
A screenshot of a Windows command prompt window. The title bar shows the path 'C:\WINDOWS\SYSTEM32\cmd' and standard window controls. The command prompt displays the output of the Java program: '101 101', '102 102', and '103 103' on three separate lines. Below this, a separator line of dashes is shown, followed by the message '(program exited with code: 0)'. At the bottom, it prompts 'Press any key to continue . . . |' with a cursor.

09.

Code.

```
class Example{
    public static void main(String args[]){
        int x=100;
        x=x++;
        System.out.println(x);
        x=x++;
        System.out.println(x);
        x=x++;
        System.out.println(x);
        x=++x;
        System.out.println(x);
        x=++x;
        System.out.println(x);
        x=++x;
        System.out.println(x);
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd.exe
100
100
100
101
102
102
103

-----
(program exited with code: 0)
Press any key to continue . . . |
```

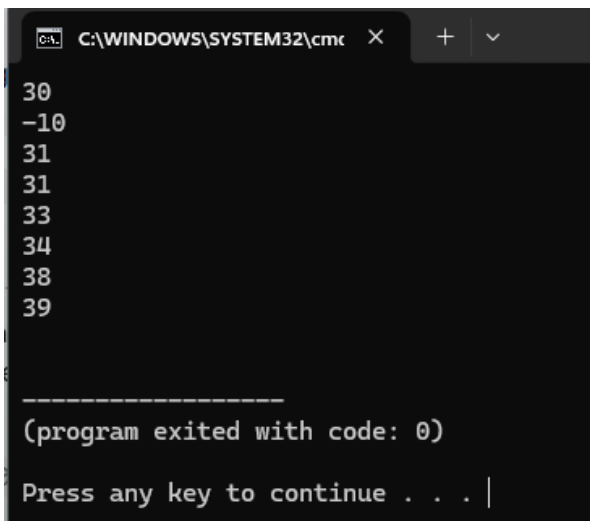
10.

Code.

```
class Example{
    public static void main(String args[]){
        int a=10,b=20;
        int x;

        x= a + b;//a-->10 + 20 = 30
        System.out.println(x);
        x= a + -b;//b-->10 + -20= -10
        System.out.println(x);
        x= ++a + b;//c--> 11 + 20 = 31
        System.out.println(x);
        x= a+ b++;//d--> 11 + 20 = 31
        System.out.println(x);
        x= ++a + b++;//e--> 12 + 21 = 33
        System.out.println(x);
        x= a++ + b++;//f--> 12 + 22 = 34
        System.out.println(x);
        x= ++a + ++b;//g--> 14 + 24 = 38
        System.out.println(x);
        x= a++ + ++b;//h--> 14 + 25 = 39
        System.out.println(x);
    }
}
```

Output.



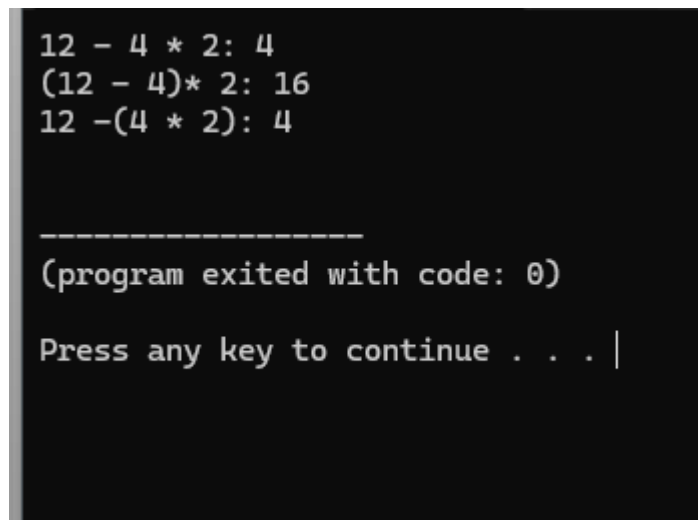
```
C:\WINDOWS\SYSTEM32\cmd.exe
30
-10
31
31
33
34
38
39
-----
(program exited with code: 0)
Press any key to continue . . . |
```

11.

Code.

```
class Example{
    public static void main(String args[]){
        int x;
        x=12 - 4*2;// line1
        System.out.println("12 - 4 * 2: "+x);//
        x=(12 - 4)*2;//line2
        System.out.println("(12 - 4)* 2: "+x);
        x=12 -(4*2);//line3
        System.out.println("12 -(4 * 2): "+x);
    }
}
```

Output.



```
12 - 4 * 2: 4
(12 - 4)* 2: 16
12 -(4 * 2): 4

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Explain -

Line1 - Multiplication (*) has a higher priority than subtraction (-).

Line2 - parentheses () are always evaluated **first**

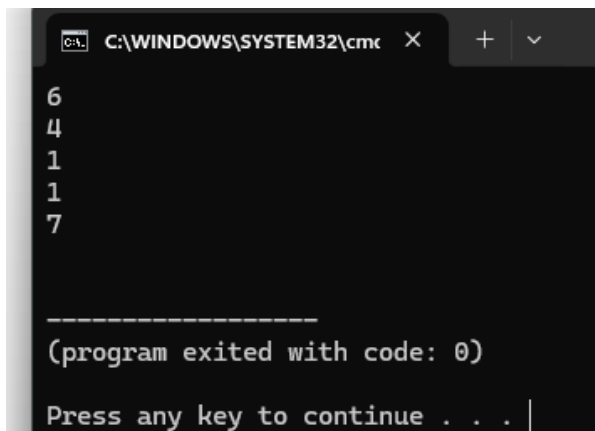
Line3 - The expression inside the parentheses is evaluated first

12.

Code.

```
class Example {
    public static void main(String args[]) {
        int x;
        x = 7 % 10 / 2 * 2; // line1 % --> / --> * (this is how its work)
        System.out.println(x);
        x = 7 % (10 / 2) * 2; //line2 (10/2) --> % --> * (this is how its work)
        System.out.println(x);
        x = 7 % 10 / (2 * 2); //line3 (2*2) --> % --> / (this is how its work)
        System.out.println(x);
        x = 7 % (10 / (2 * 2)); //line4 (2*2) --> (10/4) --> % (this is how its work)
        System.out.println(x);
        x = 7 % ((10 / 2) * 2); //line5 (10/2) --> (5*2) --> % (this is how its work)
        System.out.println(x);
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd.exe
6
4
1
1
7

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Explain.

- Line1 - $(7\%10=7 \rightarrow 7/2=3 \rightarrow 3*2=6)$
- Line2 - $(10/2=5 \rightarrow 7\%5=2 \rightarrow 2*2=4)$
- Line3 - $(2*2=4 \rightarrow 7\%10=7 \rightarrow 7/4=1)$
- Line4 - $(2*2=4 \rightarrow 10/4=2 \rightarrow 7\%2=1)$
- Line5 - $(10/2=5 \rightarrow 5*2=10 \rightarrow 7\%10=7)$

13.

Code.

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the dividend : ");
        double num1 = input.nextDouble();

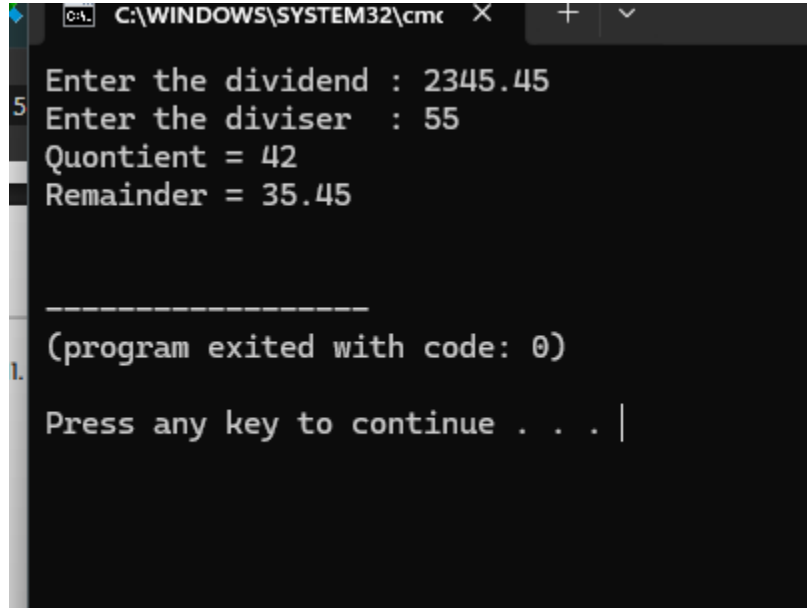
        System.out.print("Enter the diviser : ");
        double num2 = input.nextDouble();

        int quotient = (int)num1 / (int)num2;
        double remainder = num1 % num2;

        System.out.println("Quotient = "+ quotient );
        System.out.printf("Remainder = %.2f\n",remainder);

    }
}
```

output.



```
C:\WINDOWS\SYSTEM32\cmd
Enter the dividend : 2345.45
Enter the diviser : 55
Quotient = 42
Remainder = 35.45

-----
(program exited with code: 0)
Press any key to continue . . . |
```

14.

Code.

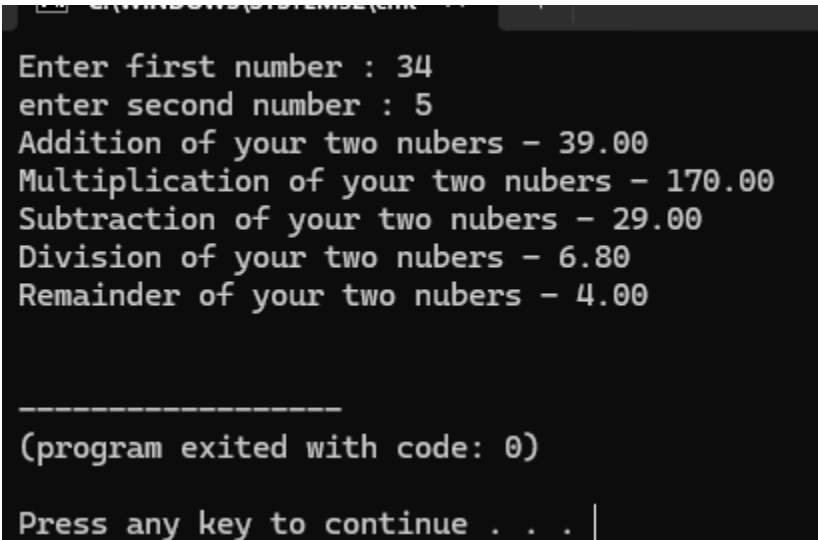
```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input =new Scanner(System.in);
        System.out.print("Enter first number : ");
        double num1 = input.nextDouble();

        System.out.print("enter second number : ");
        double num2 = input.nextDouble();

        double add = num1 + num2;
        double multi = num1*num2;
        double sub = num1-num2;
        double divi = num1/num2;
        double rem = num1 % num2;

        System.out.printf("Addition of your two nubers - %.2f\n",add);
        System.out.printf("Multiplication of your two nubers - %.2f\n",multi);
        System.out.printf("Subtraction of your two nubers - %.2f\n",sub);
        System.out.printf("Division of your two nubers - %.2f\n",divi);
        System.out.printf("Remainder of your two nubers - %.2f\n",rem);
    }
}
```

Output.



```
Enter first number : 34
enter second number : 5
Addition of your two nubers - 39.00
Multiplication of your two nubers - 170.00
Subtraction of your two nubers - 29.00
Division of your two nubers - 6.80
Remainder of your two nubers - 4.00
```

```
-----
(program exited with code: 0)
```

```
Press any key to continue . . . |
```

15.

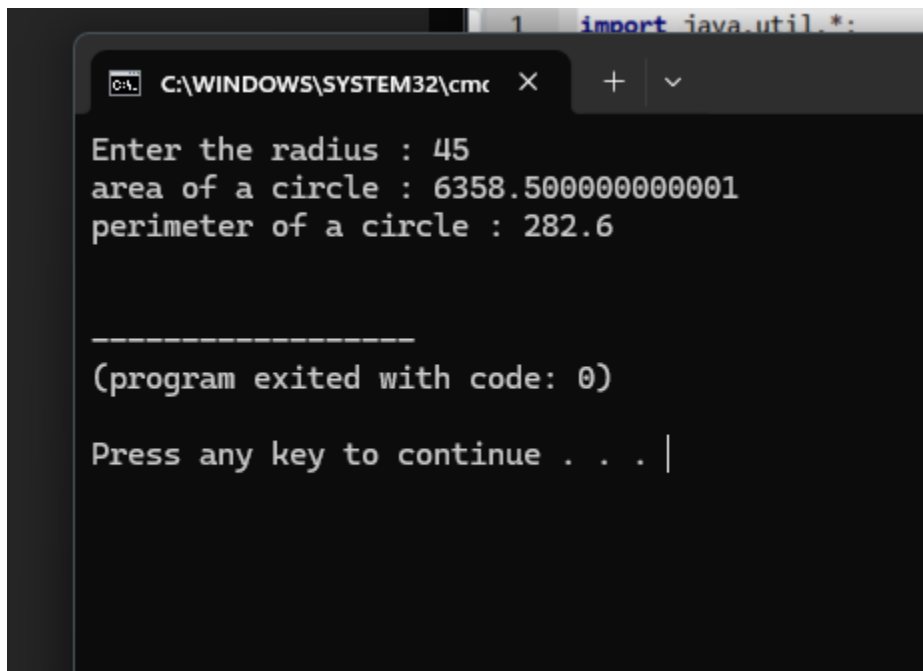
Code.

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the radius : ");
        double radi = input.nextDouble();

        double area = 3.14*radi*radi;
        double perim = 2*3.14*radi;

        System.out.println("area of a circle : "+area );
        System.out.println("perimeter of a circle : "+perim);
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter the radius : 45
area of a circle : 6358.500000000001
perimeter of a circle : 282.6

-----
(program exited with code: 0)

Press any key to continue . . . |
```

16.

Code.

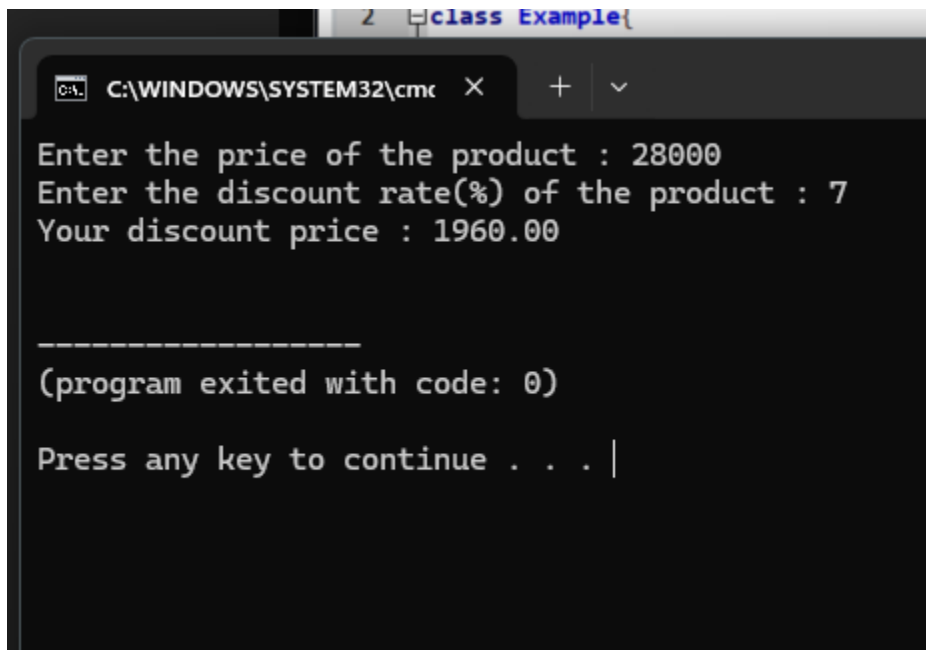
```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the price of the product : ");
        double price = input.nextDouble();

        System.out.print("Enter the discount rate(%) of the product : ");
        double dis_rate = input.nextDouble();

        double discount = price *(dis_rate/100);

        System.out.printf("Your discount price : %.2f\n",discount);
    }
}
```

Output.

A screenshot of a Windows command prompt window with a dark background. The title bar shows the file path 'C:\WINDOWS\SYSTEM32\cmd.exe'. The window contains the output of a Java program. It prompts the user to enter the price of the product, which is 28000, and the discount rate, which is 7. The program then calculates and displays the discount price as 1960.00. Below this, it shows '(program exited with code: 0)' and a prompt 'Press any key to continue . . . |' with a cursor. The code 'class Example{' is visible in the background at the top of the window.

```
class Example{
Enter the price of the product : 28000
Enter the discount rate(%) of the product : 7
Your discount price : 1960.00

-----
(program exited with code: 0)
Press any key to continue . . . |
```


17.

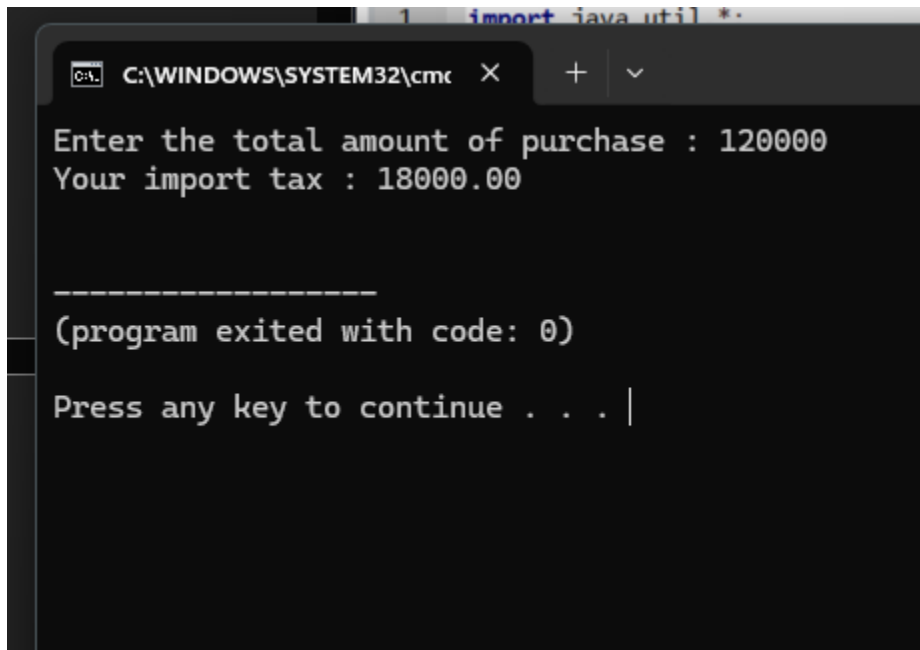
Code.

```
import java.util.*;
class Example {
    public static void main (String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the total amount of purchase : ");
        double tot = input.nextDouble();

        double tax = tot*0.15;

        System.out.printf("Your import tax : %.2f\n",tax);
    }
}
```

Output.



```
1 import java.util.*
C:\WINDOWS\SYSTEM32\cmd X + v
Enter the total amount of purchase : 120000
Your import tax : 18000.00

-----
(program exited with code: 0)
Press any key to continue . . . |
```

18.

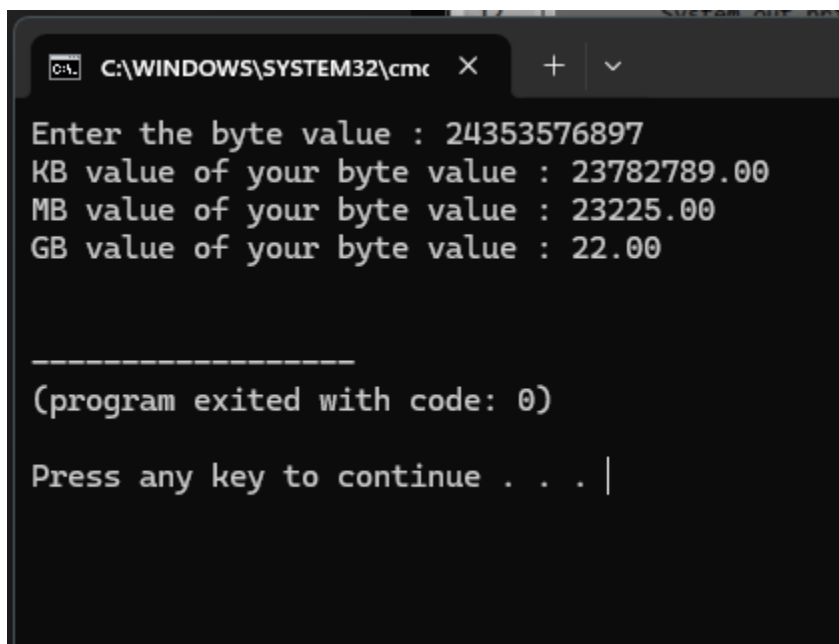
Code.

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the byte value : ");
        long b_val = input.nextLong();//byte value can be more long thats why I put the long

        double kb = b_val/1024;
        double mb = b_val/(1024*1024);
        double gb = b_val/(1024*1024*1024);

        System.out.printf("KB value of your byte value : %.2f\n", kb);
        System.out.printf("MB value of your byte value : %.2f\n", mb);
        System.out.printf("GB value of your byte value : %.2f\n", gb);
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter the byte value : 24353576897
KB value of your byte value : 23782789.00
MB value of your byte value : 23225.00
GB value of your byte value : 22.00

-----
(program exited with code: 0)
Press any key to continue . . . |
```

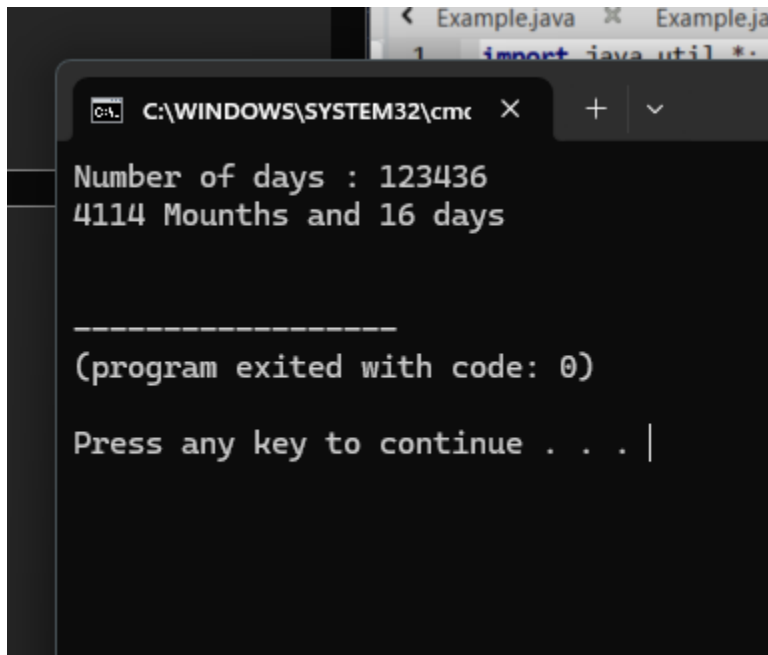
19.

Code.

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Number of days : ");
        int days = input.nextInt();

        int months = days/30;
        int rDays = days%30;
        System.out.println(months+" Mounths and "+rDays+" days");
    }
}
```

Output.

A screenshot of a Windows command prompt window. The title bar shows the path 'C:\WINDOWS\SYSTEM32\cmd.exe'. The command prompt displays the output of a Java program. It first prompts for 'Number of days' and receives the input '123436'. It then outputs '4114 Mounths and 16 days'. Below this, a separator line is shown, followed by the message '(program exited with code: 0)'. At the bottom, it says 'Press any key to continue . . . |' with a cursor. In the background, a code editor window is visible with the file 'Example.java' open, showing the first line of code: 'import java.util.*;'.

```
C:\WINDOWS\SYSTEM32\cmd.exe
Number of days : 123436
4114 Mounths and 16 days

-----
(program exited with code: 0)

Press any key to continue . . . |
```

20.

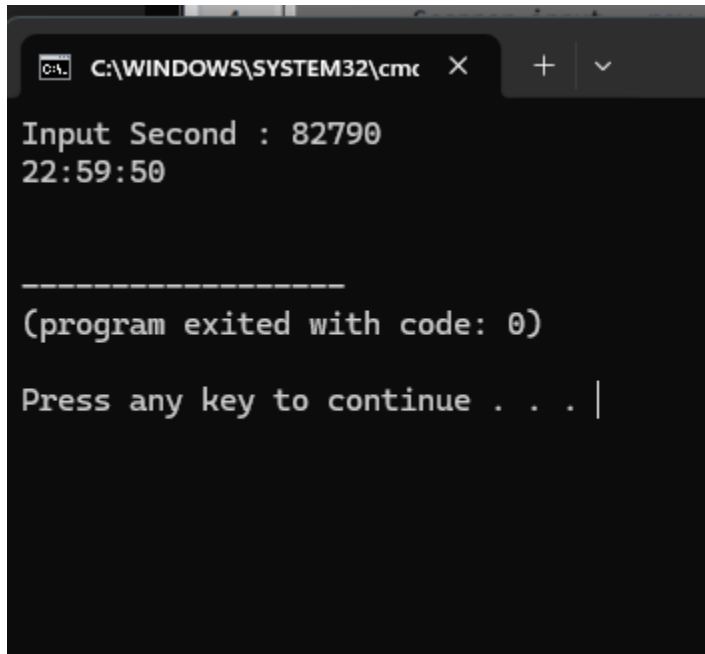
Code.

```
import java.util.*;
class Example{
    public static void main (String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Second : ");
        int sec = input.nextInt();

        int hours = sec/(60*60);
        int min = (sec%(60*60))/60;
        int fSec = sec%60;

        System.out.println(hours+":"+min+": "+fSec);
    }
}
```

Output.

A screenshot of a Windows Command Prompt window. The title bar shows the path 'C:\WINDOWS\SYSTEM32\cmd.exe'. The command prompt displays the output of the Java program: 'Input Second : 82790' followed by '22:59:50' on the next line. A horizontal dashed line separates the output from the program's exit message, which is '(program exited with code: 0)'. Below this, it says 'Press any key to continue . . . |' with a cursor at the end of the line.

```
C:\WINDOWS\SYSTEM32\cmd.exe
Input Second : 82790
22:59:50

-----
(program exited with code: 0)
Press any key to continue . . . |
```

21.

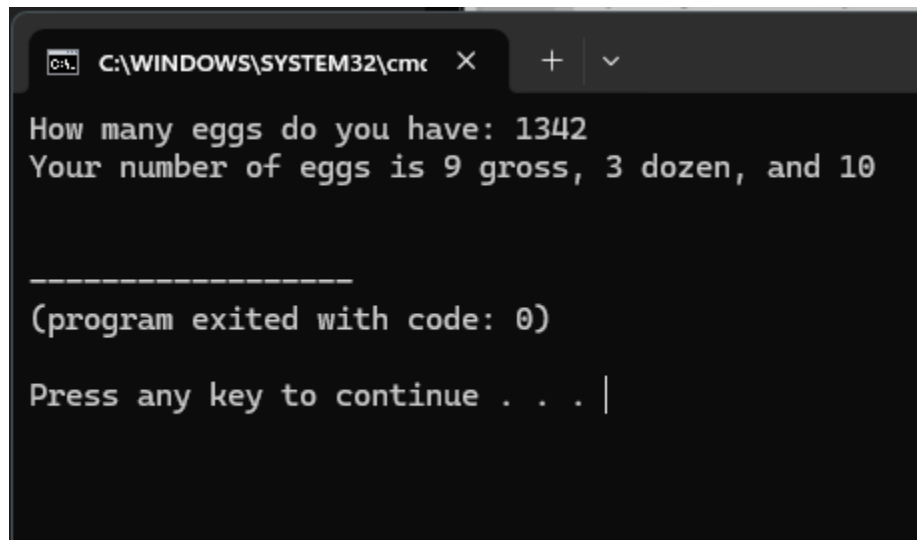
Code.

```
import java.util.*;
class Example{
    public static void main (String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("How many eggs do you have: ");
        int eggs = input.nextInt();

        int gross = eggs / 144;
        int rGross = eggs % 144;
        int dozens = rGross / 12;
        int leftovers = rGross % 12;

        System.out.printf("Your number of eggs is %d gross, %d dozen, and %d\n", gross, dozens,
leftovers);
    }
}
```

Output.

A screenshot of a Windows command prompt window. The title bar shows the file path "C:\WINDOWS\SYSTEM32\cmd.exe" with a close button. The window has a dark background with white text. The output of the program is displayed as follows: "How many eggs do you have: 1342" followed by "Your number of eggs is 9 gross, 3 dozen, and 10". There is a horizontal line of dashes below this. Then, it says "(program exited with code: 0)". At the bottom, it says "Press any key to continue . . . |" with a cursor.

```
C:\WINDOWS\SYSTEM32\cmd.exe
How many eggs do you have: 1342
Your number of eggs is 9 gross, 3 dozen, and 10

-----
(program exited with code: 0)

Press any key to continue . . . |
```

22.

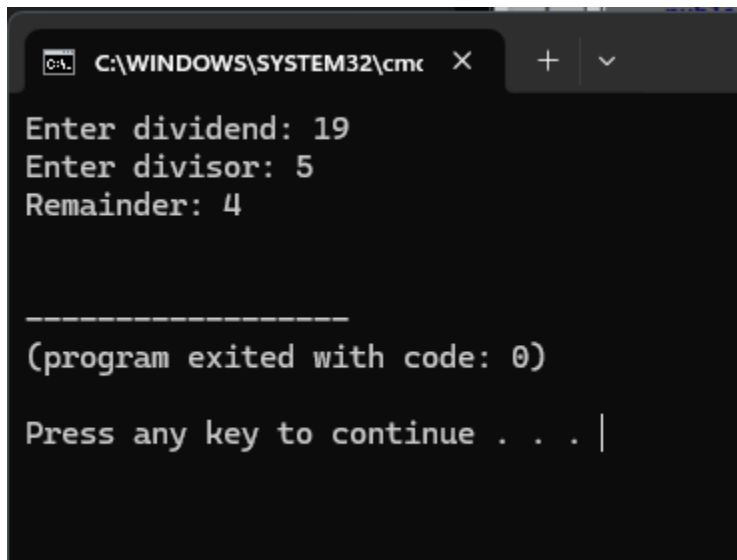
Code.

```
import java.util.Scanner;
class Example {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter dividend: ");
        int dividend = input.nextInt();

        System.out.print("Enter divisor: ");
        int divisor = input.nextInt();

        int quotient = dividend / divisor;
        int remainder = dividend - (divisor * quotient);
        System.out.println("Remainder: " + remainder);
        input.close();
    }
}
```

Output.



The screenshot shows a Windows command prompt window with the title bar 'C:\WINDOWS\SYSTEM32\cmd.exe'. The window contains the following text:

```
Enter dividend: 19
Enter divisor: 5
Remainder: 4

-----
(program exited with code: 0)

Press any key to continue . . . |
```

23.

Code.

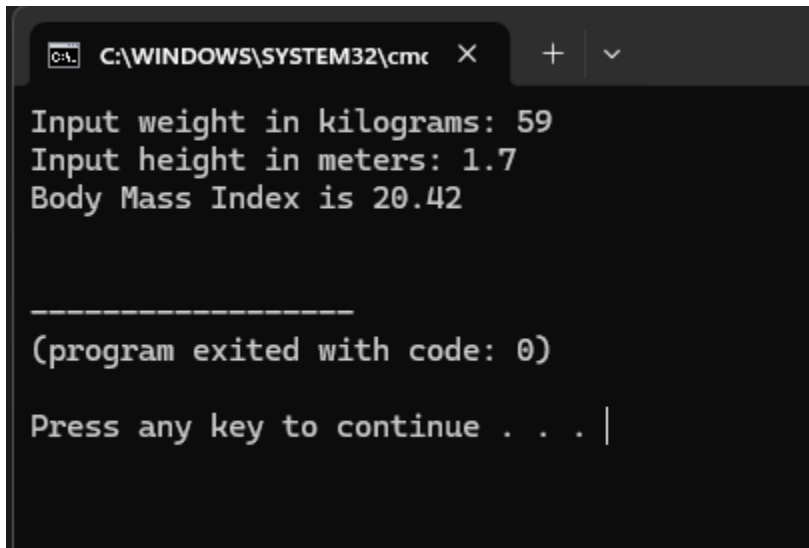
```
import java.util.Scanner;
class Example {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Input weight in kilograms: ");
        double weight = input.nextDouble();

        System.out.print("Input height in meters: ");
        double height = input.nextDouble();

        double bmi = weight / (height * height);

        System.out.printf("Body Mass Index is %.2f\n", bmi);
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd X + v
Input weight in kilograms: 59
Input height in meters: 1.7
Body Mass Index is 20.42

-----
(program exited with code: 0)
Press any key to continue . . . |
```

24.

Code.

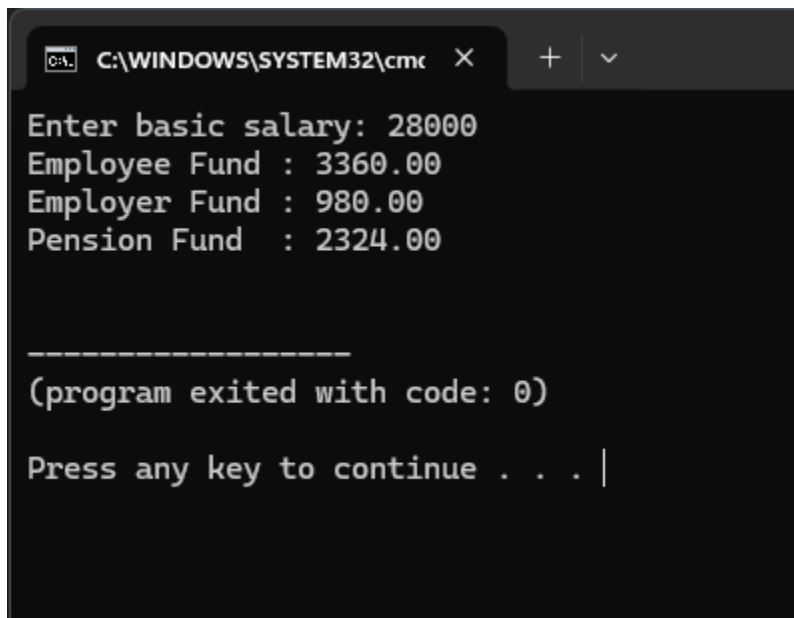
```
import java.util.Scanner;
class Example{
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter basic salary: ");
        double basicSalary = input.nextDouble();

        double employeeFund = basicSalary * 0.120;
        double employerFund = basicSalary * 0.035;
        double pensionFund = basicSalary * 0.083;

        System.out.printf("Employee Fund : %.2f\n", employeeFund);
        System.out.printf("Employer Fund : %.2f\n", employerFund);
        System.out.printf("Pension Fund : %.2f\n", pensionFund);

    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd
Enter basic salary: 28000
Employee Fund : 3360.00
Employer Fund : 980.00
Pension Fund : 2324.00

-----
(program exited with code: 0)
Press any key to continue . . . |
```


25.

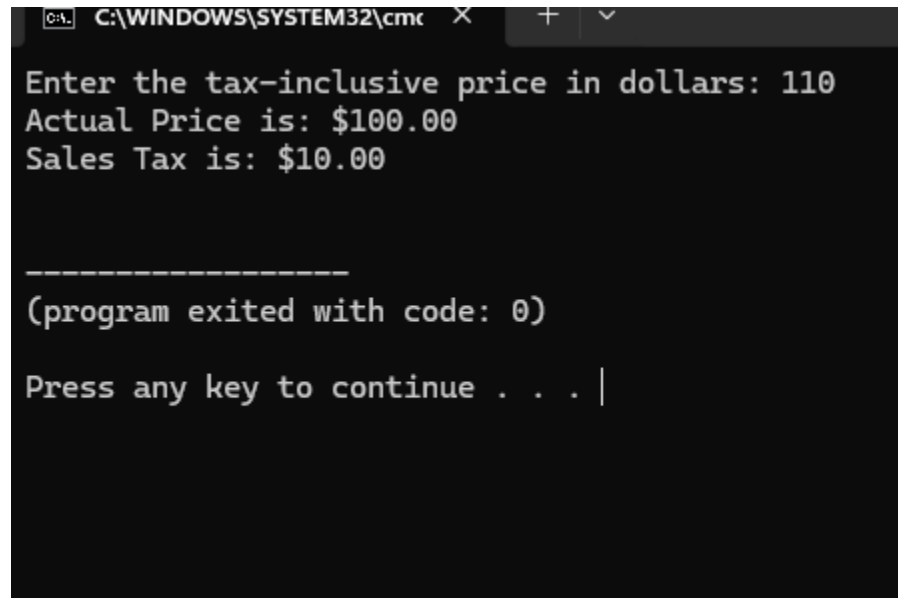
Code.

```
import java.util.Scanner;
class Example {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the tax-inclusive price in dollars: ");
        double inclusivePrice = input.nextDouble();

        double actualPrice = inclusivePrice / 1.10;
        double salesTax = inclusivePrice - actualPrice;

        System.out.printf("Actual Price is: $%.2f\n", actualPrice);
        System.out.printf("Sales Tax is: $%.2f\n", salesTax);
    }
}
```

Output.

A screenshot of a Windows command prompt window. The title bar shows the path 'C:\WINDOWS\SYSTEM32\cmd.exe'. The window has a black background with white text. The output of the Java program is displayed: 'Enter the tax-inclusive price in dollars: 110', 'Actual Price is: \$100.00', and 'Sales Tax is: \$10.00'. Below this, there is a horizontal line of dashes, followed by '(program exited with code: 0)', and then 'Press any key to continue . . . |' with a cursor at the end.

```
C:\WINDOWS\SYSTEM32\cmd.exe X + v
Enter the tax-inclusive price in dollars: 110
Actual Price is: $100.00
Sales Tax is: $10.00

-----
(program exited with code: 0)
Press any key to continue . . . |
```